

%nprocshared=64
%mem=320GB
%chk=PbRhaC12C12.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC12C12

0 1

O	-0.42135300	0.74419300	-0.52205900
O	0.58018500	2.50896100	-2.75035900
O	3.98721800	2.81101300	-2.42472900
O	1.02702400	-0.61231400	-2.46493900
C	-2.50210300	-1.85403100	-3.00494700
O	-2.85140400	-1.73695700	-1.62645600
C	-2.40754000	-0.50770500	-1.09335100
C	-0.88679900	-0.46748500	-1.07480000
C	-0.39339500	-0.57810400	-2.49455100
C	-0.97166100	-1.82371600	-3.15933900
O	3.24733300	0.99014000	-3.43625700
O	0.82523100	3.63645100	-0.86710600
C	0.21367400	3.41735400	-1.94247300
C	-0.97853300	4.28663300	-2.28202000
C	-1.96889700	4.40835400	-1.13343000
O	-2.85867000	3.25739200	-1.18578200
C	-2.83031200	5.65595700	-1.22439200
C	-3.83481000	5.77951700	-0.08765000
C	-4.65718400	7.05878700	-0.16301900
C	-5.67514800	7.17525100	0.96339500
C	-6.50386700	8.45113400	0.89852600
C	-7.52222000	8.56331000	2.02533500
C	-8.35197800	9.83873800	1.96453400
C	-3.05182200	2.53011100	-0.08506300
O	-2.39336500	2.61326200	0.92778300
C	-4.22890300	1.60320000	-0.26644500
C	-4.07978200	0.27335100	0.46422100
O	-2.81086300	-0.35815200	0.21713700
C	-3.10618500	-3.13403500	-3.52682000
O	-0.55085100	-1.91537600	-4.50812600
C	4.22140200	-1.05566600	3.44244400
C	4.70027400	-1.27139900	1.99715300
C	3.55980700	-1.83377600	1.14200100
O	2.71754800	-2.64689200	1.93078800
C	1.93600300	-1.86422700	2.84474300
C	2.74262400	-0.67652400	3.41482000
C	0.62446300	-1.42776500	2.21563400
O	2.31300800	-0.37103400	4.73045300
O	4.09186000	-2.57648900	0.10353500
O	5.02775700	-0.10027700	4.09687400
O	5.12214800	-0.01123400	1.51789200
C	3.42960100	-2.48372400	-1.16493100

C	4.49644700	-2.18594500	-2.20195300
C	2.63219100	-3.73249100	-1.50534400
C	5.04163900	-0.79269400	-2.07009400
O	5.88784200	-0.50482100	-3.04131900
O	4.72808000	-0.02119600	-1.17939500
C	6.44930900	0.83832800	-3.20482300
C	1.34554200	-3.85555300	-0.70353100
C	0.41536400	-4.94264800	-1.21978600
C	-0.96631300	-4.89064500	-0.58100900
C	-1.91117200	-5.95043700	-1.13022500
C	-3.34919400	-5.78985600	-0.65741000
C	-4.30212200	-6.78130600	-1.31141600
C	5.45477500	1.70976800	-3.98337100
C	4.13837700	1.86722300	-3.25136000
C	6.98032400	1.48241700	-1.93049800
C	7.79765800	0.54712900	-1.04740900
C	8.05467600	1.13509900	0.33522900
C	8.53876800	0.09708200	1.33752000
C	8.65431400	0.64068800	2.75418300
C	8.99657600	-0.43014100	3.78072400
C	9.06319900	0.10067100	5.20604900
C	-5.24962000	-0.65294800	0.17318300
C	-5.20949200	-1.93585900	0.98992800
C	-6.44482200	-2.80173600	0.78569900
C	-6.47152000	-4.02806300	1.68765900
C	-7.73310200	-4.86391400	1.52284000
C	-7.76055000	-6.10205000	2.40869500
C	-8.98346800	-6.97794900	2.17404200
H	-2.92357400	-0.99191100	-3.54930600
H	-2.78341200	0.30922500	-1.72653600
H	-0.53706200	-1.32523000	-0.48630800
H	-0.71851400	0.31829300	-3.04105500
H	-0.55673700	-2.70784300	-2.66794300
H	-0.58198700	5.28349500	-2.49965300
H	-1.47693800	3.92502500	-3.18280900
H	-1.43678200	4.37792200	-0.18207300
H	-2.15641500	6.52013700	-1.21729300
H	-3.34735100	5.66846000	-2.19175100
H	-4.51389400	4.91899600	-0.09969500
H	-3.30520900	5.73998100	0.87254800
H	-3.98586100	7.92717100	-0.14072600
H	-5.17589000	7.10174500	-1.12977100
H	-6.34461500	6.30510900	0.93869800
H	-5.15505500	7.12786700	1.92936900
H	-5.83475900	9.32151500	0.92453200
H	-7.02336000	8.49896700	-0.06782100
H	-8.19120100	7.69274900	1.99846800
H	-7.00220300	8.51363800	2.99136800
H	-8.87226100	9.88964000	0.99855900
H	-7.68388700	10.71007600	1.99265100

H	-5.09682600	2.12077600	0.15740100
H	-4.44460300	1.47807400	-1.32888300
H	-2.87350600	-3.25613100	-4.58555900
H	-4.19102200	-3.12059800	-3.40610200
H	-2.70530200	-3.99131100	-2.98373600
H	-0.94244100	-1.18872700	-5.00748400
H	4.32386100	-1.99017600	4.00182900
H	5.53489600	-1.98415500	1.98402900
H	2.97976600	-0.99291000	0.73520700
H	1.74210400	-2.53543600	3.68618600
H	2.62422700	0.20997400	2.77599200
H	0.01176300	-0.88625200	2.94195100
H	0.05500600	-2.29313900	1.87160300
H	0.79713100	-0.76135200	1.36718200
H	1.47685500	0.10105800	4.67845300
H	5.27458400	0.54510300	3.41950800
H	5.09275200	-0.01775400	0.54563900
H	2.73600600	-1.63930000	-1.12727400
H	5.32401000	-2.89876000	-2.12340700
H	4.09652800	-2.28563100	-3.21460100
H	2.37839100	-3.67781300	-2.57068700
H	3.26622200	-4.61846300	-1.38153800
H	1.57057600	-4.01618600	0.35303100
H	0.82671800	-2.89087500	-0.75007500
H	0.30624600	-4.84555400	-2.30921900
H	0.86566900	-5.93030900	-1.05620000
H	-0.87786500	-4.99728700	0.50796100
H	-1.40556700	-3.89802300	-0.74776100
H	-1.88905900	-5.91552900	-2.22813100
H	-1.54331000	-6.94955200	-0.86255600
H	-3.39373700	-5.89926100	0.43420000
H	-3.68886900	-4.76758100	-0.86939800
H	-3.97634900	-7.80636400	-1.08897800
H	-4.23990600	-6.67564300	-2.40311200
H	5.91860500	2.68686400	-4.13265300
H	5.26593500	1.25319100	-4.95639500
H	7.60060000	2.32479300	-2.25872500
H	6.15571600	1.90800500	-1.35921300
H	7.26264100	-0.39828800	-0.91710100
H	8.74426900	0.28876000	-1.53736400
H	8.77027200	1.96464000	0.26905700
H	7.12031700	1.56616100	0.71392200
H	7.82908500	-0.73982500	1.33899900
H	9.50393500	-0.31731300	1.01717500
H	9.40832000	1.43838700	2.78737400
H	7.70306100	1.11304900	3.03170800
H	8.24709800	-1.23103800	3.72731000
H	9.95575900	-0.89665000	3.51881700
H	8.10588000	0.57434500	5.46162700
H	9.81821800	0.89619700	5.26566800

H	-6.17220000	-0.10270700	0.39459600
H	-5.27138200	-0.88415500	-0.89784500
H	-4.30912500	-2.50372500	0.73357900
H	-5.12134400	-1.68035800	2.05429300
H	-7.34455100	-2.19940800	0.97005900
H	-6.50326000	-3.11719500	-0.26453600
H	-5.59016600	-4.64925300	1.48392300
H	-6.38099800	-3.70951200	2.73478300
H	-8.61203500	-4.24230800	1.73986000
H	-7.83335300	-5.16783800	0.47345000
H	-6.85326600	-6.69578600	2.23591700
H	-7.72228800	-5.79908000	3.46353500
H	-9.02774800	-7.25971700	1.11345900
H	-9.89390100	-6.39473900	2.36749900
H	1.35815100	-0.60439400	-3.37218300
H	-0.91148500	0.91649600	0.29436800
Pb	2.18957600	1.57273400	-1.28510800
H	-4.03298000	0.47407700	1.53751900
H	7.29051700	0.62674900	-3.86818000
C	-9.37130700	9.94971200	3.09148200
H	-10.03789500	9.07859700	3.06168100
H	-8.84994800	9.89772400	4.05561200
C	-10.19424800	11.22992600	3.02005300
H	-10.92035300	11.29257700	3.83544000
H	-9.55155100	12.11409400	3.07910400
H	-10.74758100	11.28944900	2.07740000
C	9.38197700	-0.97508400	6.23661300
H	10.33848000	-1.44818300	5.98085800
H	8.62654600	-1.76848900	6.17376200
C	9.44027100	-0.43194100	7.65885000
H	9.66911500	-1.21687700	8.38518900
H	10.20891000	0.34182200	7.75295800
H	8.48498000	0.01853200	7.94661200
C	-8.99953800	-8.23768300	3.03026500
H	-8.95624300	-7.95518600	4.08968700
H	-8.08826300	-8.81696900	2.83495800
C	-10.22523800	-9.10680200	2.77854300
H	-10.22033300	-10.00683900	3.39975900
H	-11.14741400	-8.55841600	2.99572500
H	-10.27226400	-9.42557000	1.73234900
C	-5.74630100	-6.59898700	-0.86880400
H	-5.80126300	-6.69777500	0.22055900
H	-6.06043200	-5.57237500	-1.09313200
C	-6.70583400	-7.58612100	-1.51854400
H	-6.44013700	-8.61780700	-1.26706200
H	-7.73681700	-7.42070400	-1.19302000
H	-6.68449800	-7.49790000	-2.60956000

C H O 0
6-311+g(2d,p)

Pb	0	
S	4	1.00
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2	1.00
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1	1.00
	1.6600600927	1.0000000
S	1	1.00
	0.80431655001	1.0000000
S	1	1.00
	0.22627039020	1.0000000
S	1	1.00
	0.84014530665D-01	1.0000000
P	3	1.00
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3	1.00
	2.2021426123	0.40196284806
	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1	1.00
	0.28202837058	1.0000000
P	1	1.00
	0.11333375666	1.0000000
P	1	1.00
	0.43948707430D-01	1.0000000
D	6	1.00
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1	1.00
	0.28135869813	1.0000000
D	1	1.00
	0.11500000000	1.0000000
F	1	1.00
	1.0000000	1.0000000
F	1	1.00
	0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2 3.88751200 12.20989200

2 3.81196300 16.19029100

s-f potential

4

2 12.29630300 281.28549900

2 8.63263400 62.52021700

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100

p-f potential

6

2 10.24179000 72.27689700

2 8.92417600 144.59108300

2 6.58134200 4.75869300

2 6.25540300 9.94062100

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100

d-f potential

6

2 7.75433600 35.84850700

2 7.72028100 53.72434200

2 4.97026400 10.11525600

2 4.56378900 14.83373100

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100